

Probiotics for preventing acute upper respiratory tract infections.

BACKGROUND: Probiotics may improve a person's health by regulating their immune function. Some studies show that probiotic strains can prevent respiratory infections. However, no evidence of the benefits of probiotics for acute upper respiratory tract infections (URTIs) and related potential adverse effects has been published. **OBJECTIVES:** To assess the effectiveness and safety of probiotics for preventing acute URTIs. **SEARCH STRATEGY:** We searched the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library 2011, Issue 2), which includes the Cochrane Acute Respiratory Infections Group's Specialised Register, MEDLINE (Ovid) (1950 to May week 1, 2011), EMBASE (1974 to May 2011), Web of Science which includes Science Citation Index (from 1900 to May 2011) and Conference Proceedings Citation Index (from 1991 to May 2011), the Chinese Biomedical Literature Database, which includes the China Biological Medicine Database (from 1978 to May 2011), the Chinese Medicine Popular Science Literature Database (from 2000 to May 2011) and the Masters Degree Dissertation of Beijing Union Medical College Database (from 1981 to May 2011). **SELECTION CRITERIA:** Randomised controlled trials (RCTs) comparing probiotics with placebo to prevent acute URTIs. **DATA COLLECTION AND ANALYSIS:** Two review authors independently assessed eligibility, quality of trials and extracted data. **MAIN RESULTS:** We included 14 RCTs, although we could only extract available data to meta-analyse in 10 trials which involved 3451 participants. We found that probiotics were better than placebo when measuring the number of participants experiencing episodes of acute URTI: at least one episode: odds ratio (OR) 0.58; 95% confidence interval (CI) 0.36 to 0.92; at least three episodes: OR 0.53; 95% CI 0.36 to 0.80; rate ratio of episodes of acute URTI: rate ratio 0.88; 95% CI 0.81 to 0.96; and reduced antibiotic prescription rates for acute URTIs: OR 0.67; 95% CI 0.45 to 0.98. Probiotics and placebo were similar when measuring the mean duration (MD) of an episode of acute URTI: MD -0.29; 95% CI -3.71 to 3.13 and adverse events: OR 0.92; 95% CI 0.37 to 2.28. Side effects of probiotics were minor and gastrointestinal symptoms were the most common. We found that some subgroups had a high level of heterogeneity

when conducting pooled analyses. AUTHORS' CONCLUSIONS: Probiotics were better than placebo in reducing the number of participants experiencing episodes of acute URTIs, the rate ratio of episodes of acute URTI and reducing antibiotic use. This indicates that probiotics may be more beneficial than placebo for preventing acute URTIs. However, the results have some limitations and there were no data for older people.